

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 1st Term Regular Examination, 2022
CE 3151
(Fundamentals of Transportation Engineering)

Full Marks: 210

Time: 3.00 hrs.

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

- 1 (a) Describe the relationships between landuse and transportation. Write down different landuse model. (15)
(b) Define Transportation Engineering. Explain the benefit of good system of roads. (10)
(c) What are the different modes of transportation? (10)
- 2 (a) What is traffic management? What are the objectives of traffic management? (08)
(b) Describe several ways of minimizing conflicts. Which method is suitable for Khulna city? (12)
(c) Define following terms: (15)
i) Interchange ii) Median of road iii) Shoulder
iv) Bottleneck v) Flaring
- 3 (a) What is O – D survey? What are the objectives of O – D survey? Briefly discuss the O – D survey techniques. (15)
(b) In contrast to road network in Bangladesh, discuss the road classes and responsibilities in Bangladesh. (10)
(c) Describe the advantages and limitations of Rectangular or Block pattern, Radial or Star pattern and Circular pattern. (10)
- 4 (a) What are the purposes of islands in an intersection? (10)
(b) What are the advantages of grade separation over intersection? (06)
(c) What do you mean by rotary intersection? Describe the design elements of intersection. (12)
(d) Draw a neat sketch of diamond interchange. (07)

Section – B

- 5 (a) What is Highway Engineering? Discuss the phase that dealt by Highway Engineering. (10)
(b) Derive an expression for finding overtaking sight distance at road. (10)
(c) What are the factors on which overtaking sight distance depends? Compute the minimum sight distance required to avoid head on collision of two buses approaching from the opposite directions. The speed of the buses is 60 kmph and 92 kmph respectively and each bus has brake efficiency of 70%. (15)
- 6 (a) What are the different shapes of camber? Under what circumstances each type of camber is suitable? Write disadvantages of steep cambers. (07)
(b) Define Kerb. Classify kerbs based on functions and describe briefly. (10)
(c) What is sight distance? Explain sight distance at horizontal curve and vertical summit curve – with neat sketches. (08)
(d) Determine the safe overtaking sight distance for a design speed of 85 kmp. Assume the other necessary data. (10)
- 7 (a) Define: i) Carriageway width ii) Drive ways (10)
iii) Master plan iv) Kerb v) Right of way

- (b) what is Stopping sight distance? Derive an expression for stopping sight distance on sloping ground. (12)
 - (c) A Horizontal curve is to be designed for a section of Highway having design speed of 80 mile/hour assume $f_y = 0.13$. (13)
 - i) If the physical conditions restrict the radius of the curve to 450 ft, what value is required for the super elevation at this curve?
 - ii) Is it a suitable design? why?
- 8
- (a) What is super elevation? Why is it provided? Derive an expression for super elevation on level road. (12)
 - (b) What is Transition curve? Why are transition curves provided at the beginning and end of the sharp horizontal curves? (08)
 - (c) What is PIEV? What are the purposes of PIEV? Describe the factors on which PIEV depends. (10)
 - (d) Calculate the extra widening required for a pavement of width 8.5 m on a horizontal curve of radius 220 m, if the longest wheel base of the vehicle expected on road is 5.9 m. Design speed is 85 kmph. (05)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 1st Term Regular Examination, 2022
HUM 3117
(Accounting and Auditing for Planners)

Full Marks: 210

Time: 3.00 hrs.

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

- 1 (a) Define accounting. Describe the needs of accounting in business era. (10)
(b) Define transaction. What are the features of transaction? Explain with example. (10)
(c) Describe any three principles of accounting with example and recording method. (15)
- 2 (a) What is journal? What are the difference between journal and ledger? (10)
(b) Mr. Zaman decides to open a computer programming service which names Zaman IT on January 1, 2020. During the first month of operation, the following transactions occurred. (25)

2020 January 1	Zaman invests TK 100,000 cash and equipment Tk 50,000.
January 2	Purchase computer equipment for cash Tk 20,000
January 3	Office supplies purchase on account Tk 16,000
January 5	Received cash Tk 12,000 for programming service provided.
January 6	Received a bill for Tk 25,000 from the "Daily News" for advertising on account.
January 10	Provide Tk 35,000 for programming services for customers, cash Tk 15,000 is received from customers and the balance of Tk 20,000 is billed on account.
January 15	Expenses paid in cash salaries Tk 6000, rent Tk 9000 and utilities Tk 2000.
January 17	PayTk 2500 daily needs advertising bill in cash.
January 20	The sum of Tk 6000 in cash is received from customers who have previously been billed for services in transaction 10.
January 25	Paid Tk 10,000 in cash salary expense.
Required: Prepare Journal	

- 3 (a) Describe the importance of ledger. (05)
(b) From the following transactions you are required to prepare necessary ledger accounts. (30)

2022 July 1	Mr. Reza invested cashTk 300,000 and equipment Tk 70,000 to start the agency.
July 2	Opened a bank account with cash Tk 60,000.
July 3	Office rent paid by chequeTk 10,000.
July 4	An advertisement expense occurred in the Daily Ittefaq on account Tk 5000.
July 5	Office supplies purchase in cash Tk 6000.
July 7	Advertisement bill paid by cheque transaction July 4.

- 4 (a) What do you mean by trial balance? What are the errors not detected by a trial balance? (10)
- (b) From the following ledger balances of M/S Amin & Co. You are required to prepare a trial balance as on 31st December, 2021; (25)

Cash balance	65,000 Tk	Account receivable	70,000 Tk
Accounts payable	25,000 Tk	Capital	300,000 Tk
Purchase goods	180,000 Tk	Purchase return	5,000 Tk
Transportation	10,000 Tk	Sales	395,000 Tk
Salaries expense	48,000 Tk	Sales return	4000Tk
Rent expense	60,000 Tk	Interest on Bank loan	7500 Tk
10% Bank loan	75,000 Tk	Utilities expense	1500Tk
Land	250,000 Tk	Equipment	80,000 Tk
Bad debts expense	800 Tk	Accumulated depreciation on equipment	8000Tk
Depreciation expense	4000 TK		

Section – B

- 5 (a) Define audit and auditing. How auditors detect and prevent different types of errors? Discuss with example. (15)
- (b) Discuss different types of audit with example. (10)
- (c) What is meant by 'true and fair view'? How an auditor ensure true and fair view during auditing? (10)
- 6 (a) Explain the audit process model, why an auditor should follow the model? (15)
- (b) Define audit risk. How the risk of material misstatement can be assessed? (10)
- (c) What factors an auditor should consider about the entity and its environment during auditing? Discuss briefly. (10)
- 7 (a) What is meant by internal control structure of an organization? (05)
- (b) State the significance of an adequate and effective system of internal control. (20)
- (c) Write down the features that make an internal control & system effective. (10)
- 8 (a) Discuss the provisions for reduced tax rates for export based income. (08)
- (b) What are the major sectors for tax rebate for investments or donations as per last assessment year? (07)
- (c) Calculate income tax for an individual assessee as per assessment year 2022-23 and schedule 24-A of income tax: (20)
- Mr. Shakil has the following income, deductions, and investments for the assessment year 2022-23.

Particulars	Amount (Tk)
I. Basic pay	600,000
II. House rent allowance	35% of basic
III. Medical allowance	1500/month
IV. Interest on provident fund	10,500
V. Festival allowance	100,000
VI. Bengali new year allowance	10,000
VII. Tax deducted at source	30,000
VIII. Contribution to group insurance	4,800
IX. Contribution to benevolent fund	1,800
X. Contribution to provident fund	60,000
XI. Overtime salary & bonus	150,000

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rdYear 1stTerm Examination, 2022
URP 3121
(Rural Development Planning-I)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What do you mean by development? What are the objectives of development according to Michael Todaro? (08)
- (b) "Regional economic integration, aid for trade, private sector development, and public finance are the factors responsible for boosting economic growth in a nation." Justify the statement. (12)
- (c) "Urban trap is limiting rural development", do you agree? Justify your answer. (08)
- (d) Make a comparative analysis between economic growth and economic development. (07)

2. (a) What is depreciation? Explain the internal and external causes of depreciation. (08)
- (b) Determine (i) Headcount Index (ii) Poverty Gap (PG) Index (iii) Squared Poverty Gap (SPG) Index for the following countries and explain the results. Assume poverty line is 10000. (20)

Country	Expenditures for each individual in countries					
A	9700	7500	15600	11000	7800	16300
B	15800	13200	8100	6400	9900	18100
C	5900	7800	8700	16500	12200	14700

- (c) Write the disadvantages of Headcount Index as a measure of absolute poverty. (07)
3. (a) Demarcation of rural development using GDP calculation has certain limitations. Explain. (08)
- (b) What is Human Development Index (HDI)? Calculate HDI for the following countries and explain the result. (15)

Indicators	Country P	Country Q
Life expectancy at birth (Years)	72.3	64.1
Expected years of schooling (Years)	11.2	13.8
Mean years of schooling (Years)	6.1	10.2
Gross National Income per capita (2011 PPP \$)	4057	12129

- (c) Briefly explain the drawbacks of Physical Quality of Life Index (PQLI) as a measure of development. (05)
- (d) What are the policy implications of different poverty measures? (07)
4. (a) Suppose a society consists of 5 families whose incomes are \$7000, \$14600, \$19500, \$22200, and \$34600. Construct a Lorenz curve for this five-family economy with quantile calculation. (15)
- (b) Write short note on the following: (20)
 - i. Nominal and Real GDP
 - ii. Basic Need Approach
 - iii. Poverty Line
 - iv. Absolute and Relative Poverty

Section – B

5. (a) What is rural area? (05)
(b) Differentiate rural and urban community with appropriate examples. (10)
(c) Write note on the followings: (20)
 - i) Factors influence rural settlement pattern
 - ii) Scattered and gathered rural settlements.
6. (a) Make an overview of different types of rural industries with appropriate examples. (15)
(b) Explain the process of identifying suitable rural industries by using industrial development screening matrix. (20)
7. a) Discuss raw material based different types of rural houses with appropriate examples. (20)
b) Narrate the services, utilities, infrastructure and activities of different hierarchical rural centers. (15)
8. a) Provide a brief overview of the major components of "Comilla Model" in rural development of Bangladesh. (20)
b) Write notes on the followings: (15)
 - i) Sectoral allocations in Upazila Annual Development Program
 - ii) Problems and issues in local level rural planning of Bangladesh.

Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 3rd Year 1st Term Examination, 2022.
URP 3161
(Housing and Real Estate Development)

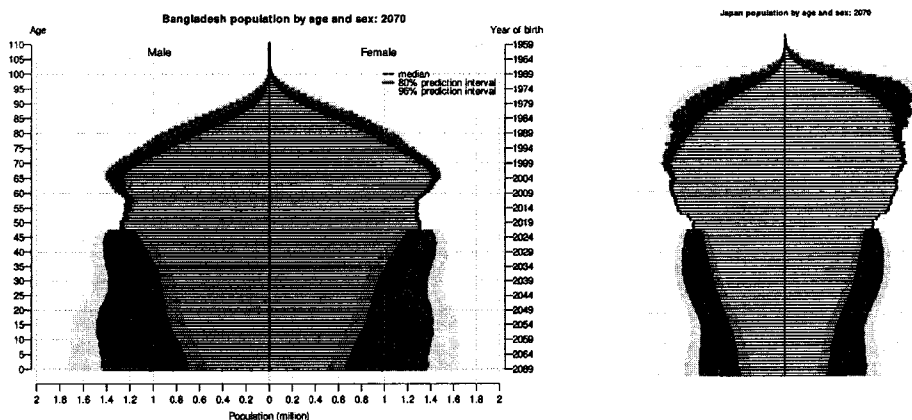
Full Marks: 210

Time: 3 hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Answer any **three** questions from each section in separate script.

Section – A

1. (a) Housing is defined as 'basic need'. Explain the difference between need and demand with example. (10)
(b) In reference to Maslow's hierarchy of needs, analyze how housing is not only a physiological need but also a need for self-esteem and self-actualization. (15)
(c) Explain the components of a functional housing. (10)
- 2 (a) Show a classification of housing based on provision with reference to Urban Bangladesh, and show a comparative analysis of different housing models. (15)
(b) What is the difference between 'Slum' and 'Squatter'? (08)
(c) Identify some major components of urban planning techniques. Explain the importance of housing as a component of urban planning. (12)
3. (a) Picture global housing need in 2100. Explain your answer from the perspective of cultural, social, demographic and technological change. (18)
(b) Following two figures show population pyramid of 2070 for Bangladesh and Japan. How will be the housing need be different for the two countries in 2070? Explain. (17)



4. Write short notes (any five): (35)
 - (a) Sites and services
 - (b) Land banking
 - (c) Hedonic pricing
 - (d) Real-Estate Investment Trust (REIT)
 - (e) Non-market factors of housing price
 - (f) Single ownership Vs Joint ownership

Section – B

5. (a) Illustrate the interaction three value determining sectors of real estate. (12)
(b) Why do we must discount future benefits of a real estate? (05)
(c) Real estate can be defined as an industry of profession – explain this statement in short. (08)
(d) Real estate tends to be highly segmented and localized – why? (10)
6. You have been appointed as an intern in East West Property Development Ltd (EWPD) of Basundhara Group. You are assigned for the following assignment.
The Padma Bridge Connects Louhajong, Munshiganj to Shariatpur and Madaripur, linking the south-west of the country, to northern and eastern regions. The company plans to invest on raw land near the Padma Brige. Now you have to do the following task.
(a) Explain the nonmonetary effects of that investment value. (10)
(b) Identify the possible investment risks. Explain the four combinations of real estate investment risk with examples. (15)
(c) Your company will purchase a parcel of land in next month for BDT 1300 million. For how much will your company have to sell the property in 30 years to earn a 10% annual return on both initial outlay and expected annual payment of BDT 75 million for property tax and development to the land? Assume these finds could be invested at comparable risk to earn a 10% annual return. (10)
7. (a) When will the downturn actually occur in a real estate market? Explain with supply and demand curve. How a developer will survive during a market downturn? (15)
(b) What is real estate valuation? Summarize the primary principle of real estate valuation. (20)
8. Write short notes on any five of the following: (35)
 - (a) Market research
 - (b) Economic base of a city
 - (c) Supply side of urban growth
 - (d) Bid-rent model
 - (e) Adjustment to comparable property transaction prices
 - (f) Reproduction and replacement cost
 - (g) Accrued depreciation

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 1st Term Regular Examination, 2022
CE 3107
(Elements of Solid Mechanics)

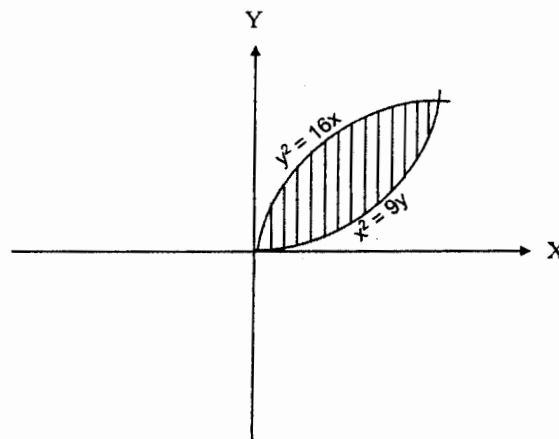
Full Marks: 210

Time: 3.00 hrs

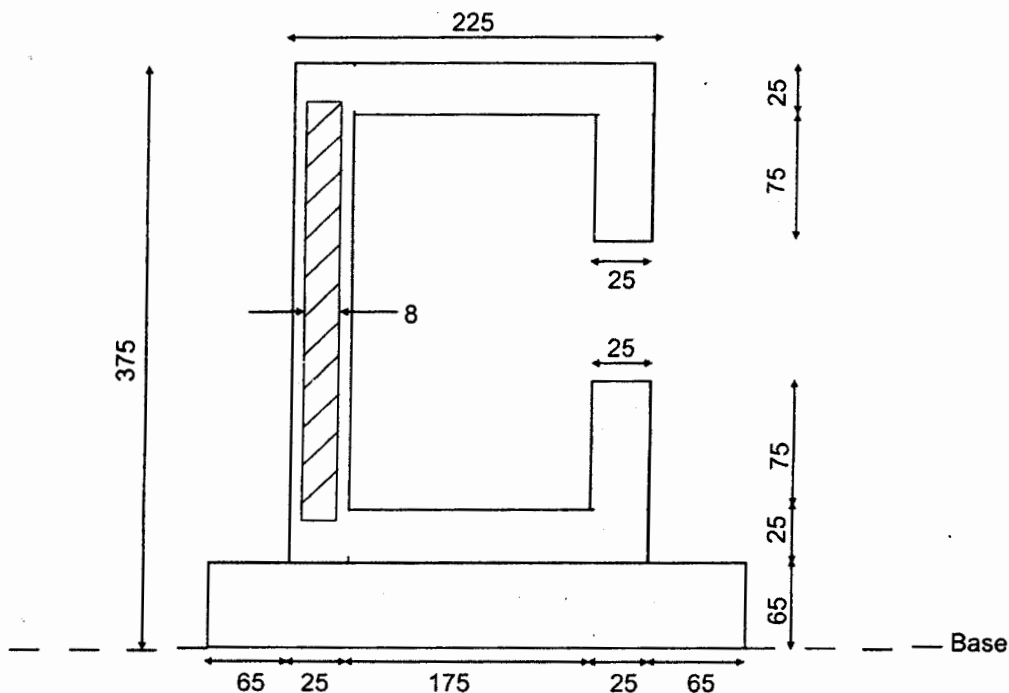
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Section – A

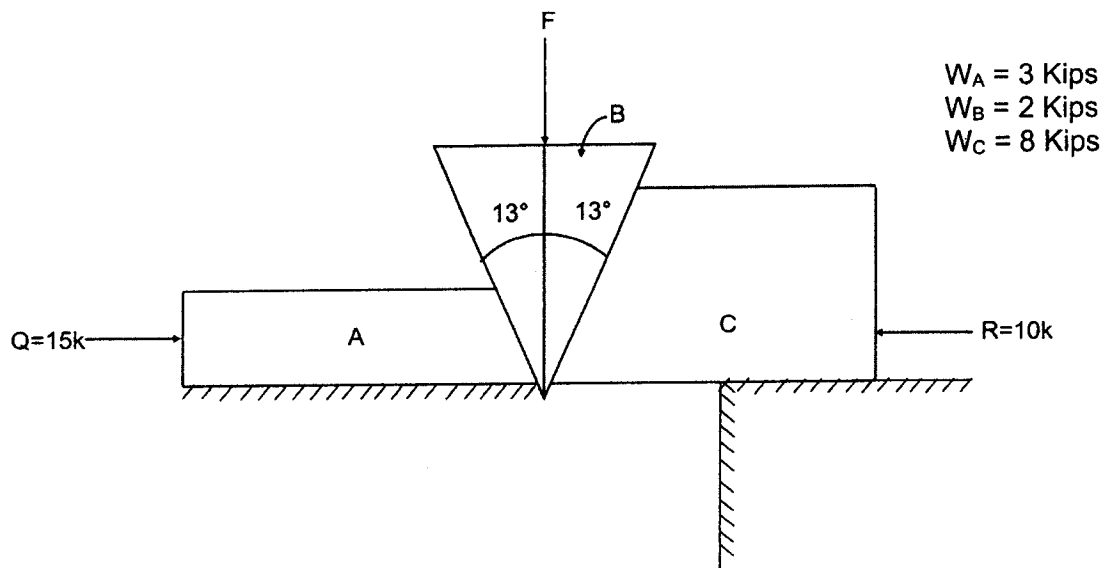
- 1 (a) Differentiate between center of mass and centroid. Derive the formula for centroid of triangles. (15)
- (b) Determine the location of the centroid of the area enclosed by the arcs of the parabolas, $y^2 = 16x$ and $x^2 = 9y$, where the linear units are inches. (20)



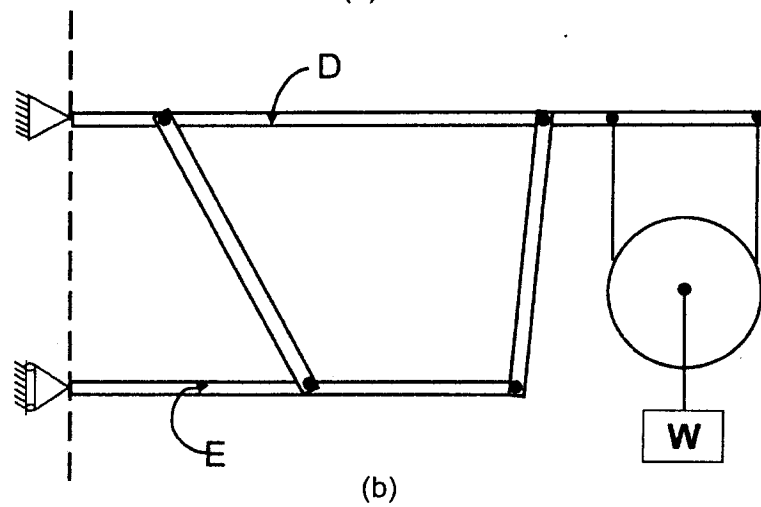
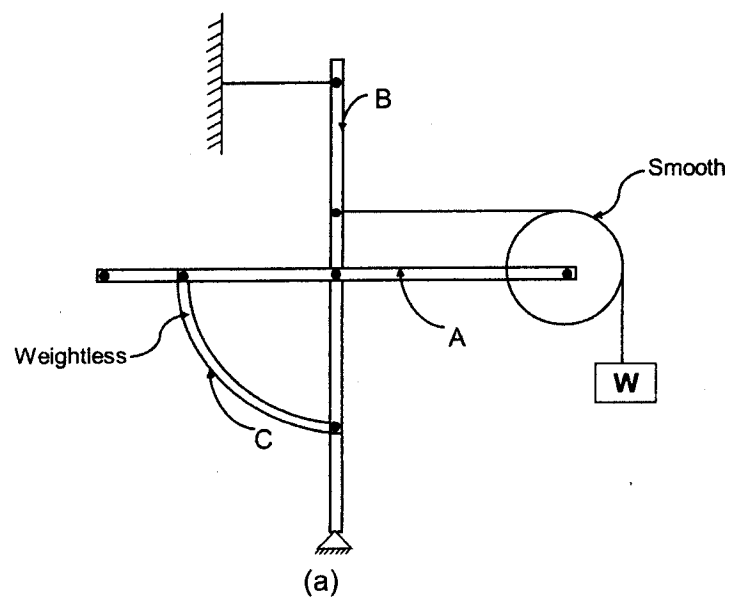
- 2 (a) Prove that $I_x = \bar{I}_x + Ad^2$, where the symbols bear their usual meanings. (10)
- (b) Determine the moment of inertia of the unshaded area of the given section below with respect to the base and the centroidal axis parallel to the base. All dimensions are in mm. (25)



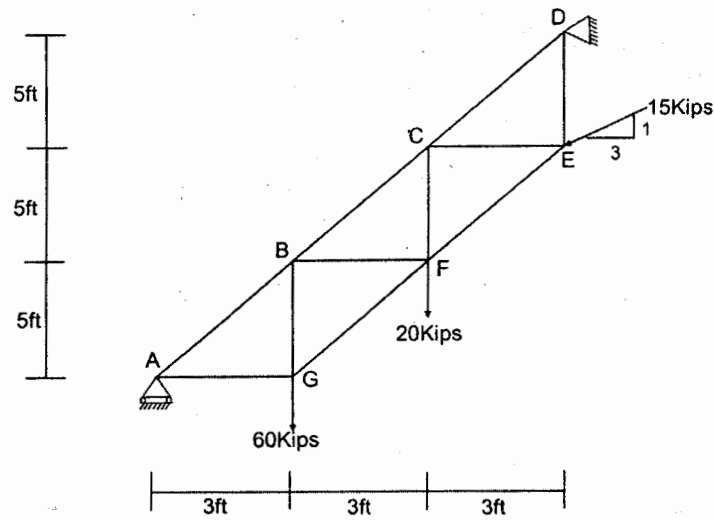
- 3 (a) Describe transmissibility of a force. (8)
- (b) In the following figure, neglecting all friction so that the reactions are normal to the surfaces, find the force F on top of the wedge. Assume that each body is subjected to a concurrent force. (17)



(c) Draw free body diagram of members A, B, C, D, and E of the following figures. (10)

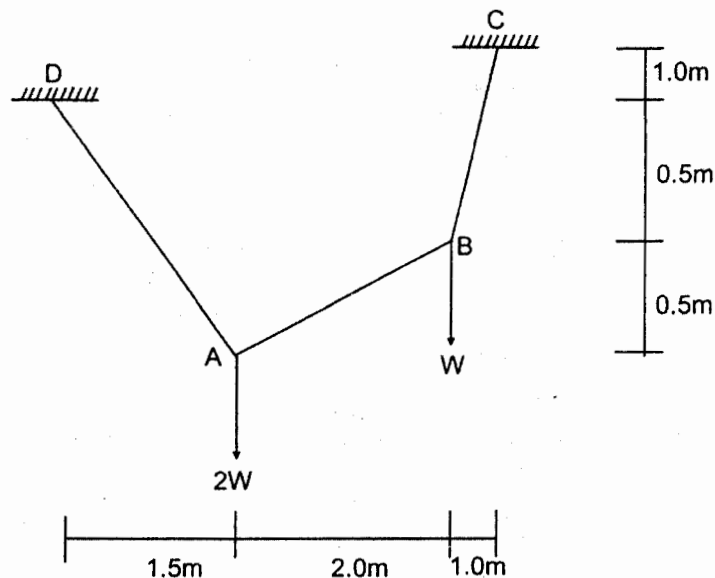


- 4 (a) Define i) zero force member ii) truss (04)
- (b) Determine the cross-sectional areas of member AG, CF, BF, and CE for the truss shown in figure. The stresses are not to exceed 25ksi in tension or 12ksi in compression. (31)



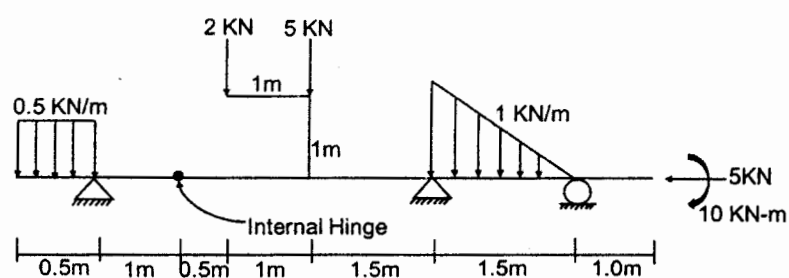
Section – B

- 5 (a) Define i) Engineering stress, ii) Modulus of Rupture, iii) Necking (06)
- (b) Draw the stress-strain diagram of mild steel and show all the stress limit. (08)
- (c) For a given structure shown in the figure, determine the safe load 'W' that can be applied. (21)

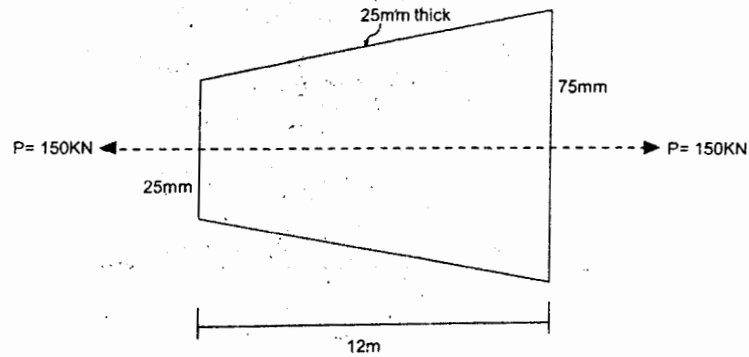


Member	Diameter(mm)	Maximum allowable axial stress (MPa)
AD	80	100
AB	72	70
BC	55	50

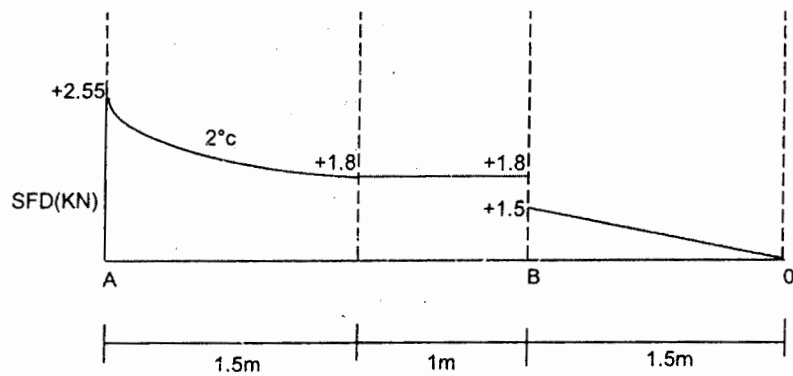
- 6 (a) Define i) Point of Contraflexure, ii) Dangerous section, iii) Modulus of elasticity (06)
- (b) Draw the shear force and Bending moment diagram for the structure shown below and identify dangerous section. (29)



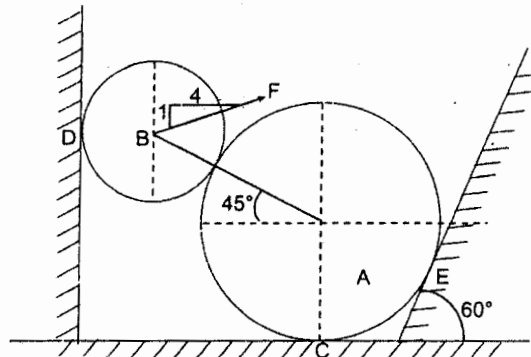
- 7 (a) Compute the total elongation caused by an axial load of 150kN applied to a flat (15)
bar 25mm thick, tapering from a width of 150mm to 50mm in a length of 12m as
shown in the figure. Assume, $E = 205\text{GPa}$.



- (b) Draw the moment and load diagram from the following shear diagram. Specify (20)
values at all changes of load positions and at all points of zero shear [A and B
are the support for the structure].



- 8 (a) Two spheres are at rest against smooth surfaces, as shown below. Sphere A (20)
weights 4000lb and sphere B weights 1200lb. Let, $F = 1300\text{lb}$. Find the reaction at
C, D, and E.



- (b) Find the value of F and θ for the following force system in the figure below. Also (15)
find the resultant of the force system.

